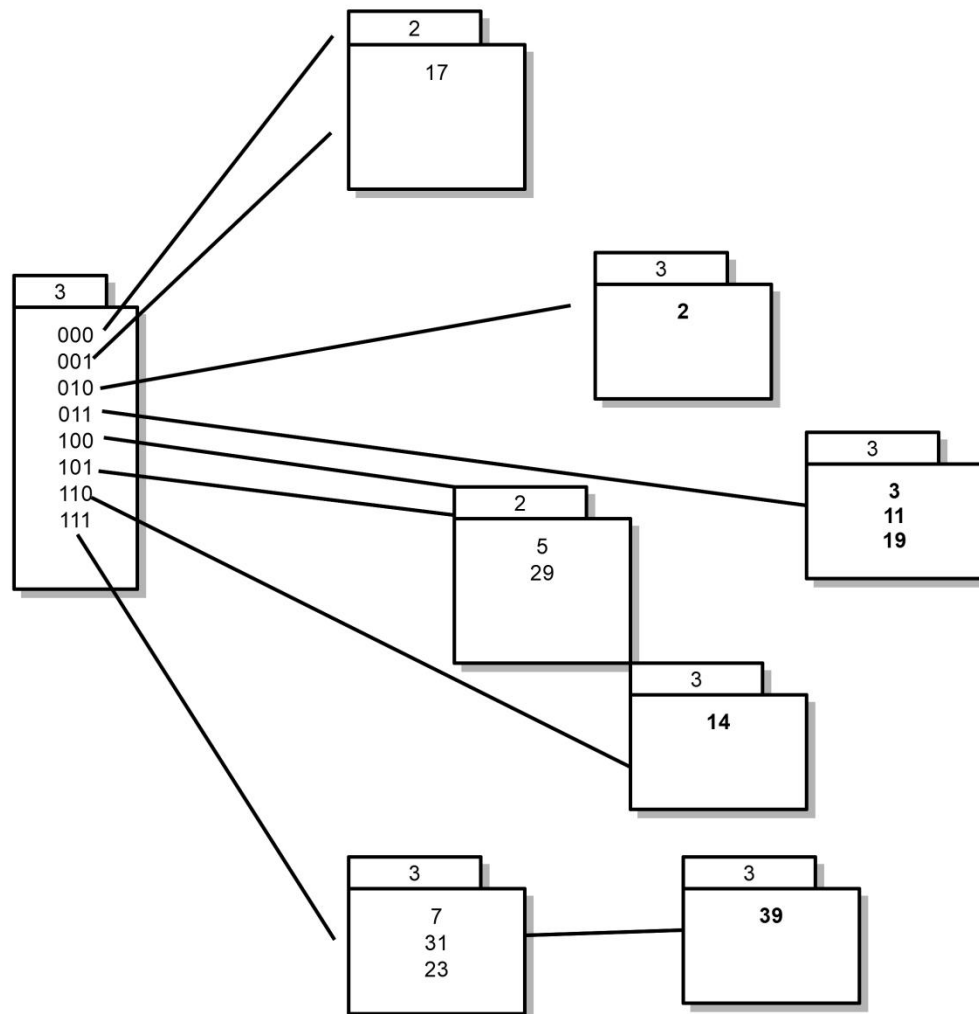


Part1
Problem1



Initial condition:

000, 001 → Bucket A (ld=2): {17}

010 → Bucket B (ld=3): {2}

011 → Bucket C (ld=3): {3, 11, 19}

100, 101 → Bucket D (ld=2): {5, 29}

110 → Bucket E (ld=3): {14}

111 → Bucket F (ld=3): {7, 31, 23} + overflow {39}

(1) Delete 11:

$h(11) = 3 \rightarrow 011 \rightarrow \{3, 11, 19\} \rightarrow \text{remove} \rightarrow \{3, 19\}$

We can merge {3, 19} and {2} and the local depth becomes 2. State after (1):

Directory	LD	Contents
000, 001	2	{17}
010, 011	2	{2, 3, 19}
100, 101	2	{5, 29}
110	3	{14}

111	3	{7, 31, 23} + overflow {39}
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(2)

$h(31) = 7 \rightarrow 111 \rightarrow$ main bucket {7, 31, 23}, overflow {39}

Remove 31 $\rightarrow$ main: {7, 23}, overflow: {39} $\rightarrow$ total = 3 $\leq$ capacity $\rightarrow$ merge overflow back

Bucket 111 = {7, 23, 39}, overflow eliminated.

Check further merge: buddy of 111 is 110 = {14}, combined = 4 $>$ 3, cannot merge.

State after (2):

Directory	LD	Contents
000, 001	2	{17}
010, 011	2	{2, 3, 19}
100, 101	2	{5, 29}
110	3	{14}
111	3	{7, 23, 39}

(3) $h(1) = 1 \rightarrow$ last 3 bits = 001 $\rightarrow$ Bucket {17}, not full $\rightarrow$ insert directly

State after (3):

Directory	LD	Contents
000, 001	2	{17, 1}
010, 011	2	{2, 3, 19}
100, 101	2	{5, 29}
110	3	{14}
111	3	{7, 23, 39}

(4) $h(15) = 7 \rightarrow 111 \rightarrow$ {7, 23, 39}, full

$ld = 3 = GD = 3 \rightarrow$ overflow

State after (4):

Directory	LD	Contents
000, 001	2	{17, 1}
010, 011	2	{2, 3, 19}
100, 101	2	{5, 29}
110	3	{14}
111	3	{7, 23, 39}+overflow {15}

Problem2

(1) Binary search repeatedly cuts the search space in half until the target block is found.

The maximum number of block accesses is: $\lceil \log_2 (10,000) \rceil \approx \lceil 13.29 \rceil \rightarrow = 14$

because:

$$2^{13} = 8192 < 10000, 2^{14} = 16384 > 10000$$

So, in the worst case, binary search needs 14 block accesses to locate the block containing the search value.

(2) If the attribute is not a key, multiple records may have the same value. Even after binary search locates the first matching block, we must continue scanning to find all matching records. Since the file is sorted, matches will be contiguous, but we don't know how many there are ahead of time.

In the worst case, all records could match, requiring reading all 10,000 blocks. More precisely:

Binary search still requires $\lceil \log_2 (10,000) \rceil = 14$ blocks to find the first matching record.

After that, we must scan forward (and possibly backward) to retrieve all matching records.

Worst case: all 10,000 blocks are accessed.

Problem3

(1) Number of runs = $\lceil br/M \rceil = \lceil 20,000/5 \rceil = \lceil 4,000 \rceil = 4,000$ runs

(2) The total merge passes = $\lceil \log_{M-1}(\lceil br/M \rceil) \rceil = \lceil \log_4(4000) \rceil = \lceil 5.98 \rceil = 6$

(3) The total number of blocks transfers = $2 * br * \lceil \log_{M-1}(\lceil br/M \rceil) \rceil + 2 * br = 2 \times 20000 \times (6+1) = 280000$ blocks

(4) Total time = $280000 \times 15\text{msec} = 4,200,000 \text{ ms} = 4,200 \text{ seconds} = 70 \text{ minutes}$

(5) Two passes = 1 run-creating pass + 1 merge pass

After the run-creating pass using M buffer pages, we produce: runs = $\lceil 20,000/M \rceil$

For one merge pass to handle all runs, we need enough input buffers: number of runs $\leq M-1$

or number of runs $< M$

So we need: $\lceil 20,000/M \rceil \leq M-1$, then $20,000/M \leq M-1$, $M(M-1) \geq 20000$

Testing:

$$141 \times 140 = 19740 < 20000$$

$$142 \times 141 = 20022 \geq 20000$$

So, the minimum number of buffer pages required is 142.

Problem4

$br = 20000/25 = 800$ blocks

$bs = 45000/30 = 1500$ blocks

Cost = $br * bs + br = 800 \times 1500 + 800 = 1200800$ block assesses or $br * bs + bs = 800 \times 1500 + 1500 = 1201500$ block assesses

Minimum cost uses R as outer: 1200800 block accesses.